



Clashing AIs: Artificial Intelligence Literacy and Academic Integrity in an EFL Context

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Abstract

Using a quantitative correlational approach, this study examined the connection between artificial intelligence (AI) literacy and attitudes towards plagiarism among English Language Teaching (ELT) students in a Turkish higher education context, where English as a foreign language (EFL) is taught. The participants ($N=82$) scored high levels of AI literacy, especially in terms of usage, but knowledge of the ethical components, such as data privacy, responsible use, and authorship, was minimal. Their plagiarism attitudes were moderate and nuanced, marked by ethical awareness on one hand and tolerance for self-plagiarism acts on the other. The findings showed no significant relationship between AI literacy and plagiarism attitudes, suggesting that merely using AI does not lead to heightened ethical awareness. Furthermore, no significant differences were observed across gender, academic year, or frequency of AI use, indicating that these factors may be less influential than shared institutional experiences. Lastly, despite its small sample, the study contributes valuable insights into the foundational challenges of ethics integration in AI literacy, particularly for ELT students as future educators. The findings highlight the need for teacher education programs to embed the dimensions of ethical and critical use of AI tools into curricula to prepare students for responsible digital citizenship.

Keywords AI ethics · AI literacy · AI plagiarism · EFL students · Teacher education

Introduction

Artificial intelligence (AI) technologies have been growingly used in every domain, with education being a no exception, as Zhang and Aslan (2021) highlight. AI has the potential to revolutionize education through such features as course design, assessment, personalized education, immediate corrective feedback, and ease of access to information, thereby increasing student engagement and motivation (Wei, 2023). These features are particularly

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important in language education, which in this study pertains to the teaching of English as a foreign language (EFL).

Among AI technologies, large language models (LLMs) such as ChatGPT, Gemini, and Claude have become popular tools for academic writing. These generative AI (GenAI) tools can produce coherent and contextually appropriate text from simple prompts. This makes them especially attractive to EFL learners who seek efficiency and writing support. Research highlights several pedagogical benefits, including automated feedback, enhanced linguistic accuracy, and improved learner motivation and self-efficacy (Lo et al., 2024; Teng, 2024; Li et al., 2024; Woo et al., 2025).

However, the increasing use of GenAI also raises important concerns about academic integrity. This is particularly relevant in English Language Teaching (ELT) programs, where preservice teachers may struggle with original writing and citation. Some students may submit unedited AI-generated texts, which challenges legitimate authorship, and existing detection tools may fail to flag such outputs effectively (Eke, 2023; Barrett & Pack, 2023; Sullivan et al., 2023). These issues point to the need for stronger AI literacy, institutional policies, and ethical awareness in academic writing.

In addressing these issues, understanding AI becomes critical. An individual's level of familiarity and knowledge influences how they conceptualize, understand, and utilize it (Rosling & Littlemore, 2011; Wang et al., 2023). Thus, the current study aims to explore the relationship between AI literacy and academic integrity perceptions among ELT students. Understanding this relationship could help policymakers and program developers design interventions that promote responsible AI use while safeguarding academic integrity in language education. It can also provide insights into how well future educators are prepared to navigate AI-related ethical challenges.

Literature Review

Academic Integrity Problems

GenAI tools have introduced new challenges to academic integrity in higher education. Students may now use AI to generate assignments and submit them as their own, a practice described as AI plagiarism, which undermines honesty and learning (White, 2023). This bypasses the learning process and violates expectations of student authorship since the cognitive ideas, analytical processes, and words produced belong to the AI tool, not the student (Bhatia & Kulkarni, 2023; Cotton et al., 2023). In educational contexts, authorship entails intellectual contribution and responsibility, which AI cannot fulfil (Thorp, 2023). Furthermore, a major concern here is that AI-generated content is technically original and not copied from existing sources. This makes it undetectable by traditional plagiarism software and harder for instructors to identify academic misconduct (Perkins, 2023). Consequently, when students submit unattributed AI-generated content, they misrepresent authorship and compromise originality, even if the text is novel and grammatically sound (Eke, 2023). The issue is particularly pressing in EFL contexts, where learners may resort to AI to overcome language barriers or meet deadlines.

In light of these concerns and AI's growing presence in academia, the discourse has moved away from outright bans, which are mostly seen as impractical and ineffective (Eke,

2023; Kirwan, 2023). Instead, recent efforts emphasize ethical and responsible use, supported by institutional frameworks and AI literacy initiatives. Therefore, many universities and instructors now focus on defining acceptable uses of AI, often through clear guidelines. For instance, students may be permitted to use AI for brainstorming or revision, but not for writing entire assignments or interpreting experimental results. Students may also be required to disclose if and how AI was used. Eke (2023) recommends that users take responsibility for transparently reporting AI involvement to uphold academic integrity. He recommends documenting the generation date, the prompt used, and limiting direct quotations to clearly marked sections until more robust detection tools are available. Similarly, educators emphasize the importance of developing AI literacy, understanding both the capabilities and limitations of AI, and learning how to ethically incorporate it into writing for educational purposes (Cotton et al., 2023). Thus, promoting awareness of AI as a supportive tool rather than a shortcut can help students improve their writing skills without compromising academic integrity.

AI Literacy

AI literacy includes the knowledge and skills needed to understand AI systems and how to interact with them, which requires a critical understanding of how they function. Chiu et al. (2024) define AI literacy in three domains: understanding the logic of AI, responsible use of AI, and working and collaborating with or adapting to AI tools. It also encompasses a range of skills, including technical skills, critical thinking, and ethical reasoning (Chiu et al., 2024), which support the user's ability to interpret AI output, identify bias or inaccuracies, and use AI technologies ethically. These skills go beyond knowing how to use the technology, requiring a deeper and reflective understanding of how AI is involved in learning, creativity, and evaluative functions. Kovári (2024) proposes, AI literate learners can evaluate the value of the AI-generated content and use it without overly depending on the AI to create their product. Research by Hossain et al. (2025) supports this, showing that EFL students with strong AI literacy can use AI responsibly in their writing by employing AI suggestions while clearly distinguishing their own ideas and avoiding plagiarism. Therefore, developing AI literacy among EFL learners is essential to ensure the ethical and effective use of GenAI tools in academic writing.

AI literacy thus stands critical in addressing the ethical concerns related to generative AI use in academic writing. As stated by Hossain et al. (2025), many EFL learners rely on AI for surface-level tasks ranging from basic grammar corrections to translations because they do not fully understand how these tools function. This surface-level engagement can lead to dependence on the output and, potentially, to academic dishonesty. However, once learners are shown the acceptable and ethical parameters of AI use, they are academically more responsible (Kovári, 2024). For example, they should know when to disclose they have used AI support or to not allow AI to create an entire essay. In short, as stated by Chiu et al. (2024), if AI literacy is instilled, students could be equipped to adapt to a variety of ethical issues stemming from AI, including authorship, plagiarism, and potential biases.

Relevant Research

Existing research highlights increasing concerns about academic integrity in higher education due to the widespread use of AI writing tools. While such tools can improve fluency and writing efficiency, they also pose risks of plagiarism and misinformation. Perkins (2023) notes that AI-generated content may evade traditional plagiarism detectors, and Bittle and El-Gayar (2025) emphasize concerns over misattribution of authorship despite the instructional potential of generative AI. Even before ChatGPT, tools like Google Translate and Grammarly raised concerns regarding originality when used without critical engagement (Gustilo et al., 2024). Now, there is a growing need for institutional policies, AI detection strategies, and ethical frameworks to respond to the changing paradigm in academic writing, supported by AI literacy (Bittle & El-Gayar, 2025; Perkins, 2023).

However, research specifically focusing on EFL contexts remains scarce. In a study by Hossain et al. (2025), they observed that while Turkish EFL students often used AI tools in writing, they hesitated to frame this in ethical terms. Similarly, Khasawneh (2024) found that EFL preservice teachers with stronger academic integrity beliefs were less inclined to use ChatGPT for assignments. Werdiningsiha et al. (2024) reported that EFL students used ChatGPT for vocabulary support while emphasizing the importance of maintaining authorial authenticity. Despite these insights, studies directly examining how future language educators navigate GenAI use and academic integrity frameworks are still limited, a gap the present study aims to address.

Research Questions

In light of the research purpose, the following research questions were formulated.

RQ1: What are the AI literacy levels of Turkish ELT students?

RQ2: What are the attitudes of Turkish ELT students toward plagiarism?

RQ3: Do demographic variables significantly affect AI literacy and attitudes toward plagiarism?

RQ4: Is there a significant correlation between AI literacy and attitudes toward plagiarism?

Method

Research Design

This study employed a correlational research design to examine the possible relationship between AI literacy and the participants' attitudes toward plagiarism. Correlational research entails examining potential relationships between two or more variables within the same sample group or between the same variables across different sample groups (Leedy & Ormrod, 2010). In the case of the present study, the correlation between two variables, AI literacy and attitudes toward plagiarism among ELT students were taken into account. Additionally, the study also involved a comparative aspect, analysing differences, if any, among

the sampling groups in their AI literacy levels and perceptions of plagiarism depending on the demographic variables.

Setting and Participants

The research was carried out within the ELT department at a higher education institution in Turkey. It employed convenient sampling, which is the selection of participants from preestablished groups (Etikan et al., 2015). A number of 82 ELT students were gathered, consisting of 37 s-year and 45 fourth-year students. In terms of gender disparity, 44 of the participants were male while the remaining 38 were female. The ages of the participants varied from 19 to 28 with an average of 22. The sample was appropriate for exploring the relationship between AI literacy and students' attitudes towards plagiarism because all participants had relevant experience using AI tools for educational and course-related purposes. Further details about participant demographics are available in Table 1. As demonstrated by the table, the ages were grouped purposely so that age balances were observed across categories. Data demonstrated that AI tools were significantly used among all participants, with the majority reporting often (54.9%) AI tool usage, followed by always (22.0%). Correspondingly, none of the participants reported never using an AI tool, denoting a definite presence of AI in their academic routines.

Data Collection

Prior to data collection, an institutional permission was acquired from the faculty. Then, the study and its significance were explained to the students. The surveys were administered to those who were willing to participate, both online through Google Forms and face-to-face, based on the desired number of participants. Demographic information was also gathered.

Data were collected with two validated survey instruments. The first one was by Wang et al. (2023) to measure AI literacy. Constructed with a 5-point Likert-type scale, the AI literacy scale (AILES) consists of 12 items under four subscales with 3 items each: awareness, usage, evaluation, and ethics. As for the reliability scores of the instrument, AILES demonstrates an acceptable level of Cronbach's alpha ($\alpha=0.83$), and the subscales also exhibit reliability levels of more than 0.70. AILES's validity was first established in terms of content by expert review, and then, validity analyses were done according to Larcker (1981), who suggest that composite reliability (CR=0.88, overall) and average variance extracted (AVE=0.50, overall) scores be considered.

The second research instrument was by Mavrinac et al. (2010) to measure the attitudes on plagiarism, entitled the Attitudes Toward Plagiarism Scale (ATPS). The instrument

Table 1 Age distribution and AI use frequency ($N=82$)

Category	Variable	<i>n</i> (%)
Age group	19–20 years	25 (30.5%)
	21–22 years	31 (37.8%)
	23–28 years	26 (31.7%)
Frequency of AI use	Never	0 (0%)
	Rarely	2 (2.4%)
	Sometimes	17 (20.7%)
	Often	45 (54.9%)
	Always	18 (22.0%)

has a 5-point Likert-type scale and includes 29 items under three subscales: positive attitude toward plagiarism (12 items, $\alpha=0.83$), negative attitude toward plagiarism (7 items, $\alpha=0.79$), and subjective norms toward plagiarism (7 items, $\alpha=0.85$). The reliability scores are beyond the acceptable threshold levels. As for validity, ATPS' content validity was established with a panel of three experts, and then, two rounds of principal component analysis were used for construct validation. Each item had a factor loading of more than 0.35 under their pertaining subscales, with eigenvalues of 9.18, 1.94, and 1.47 for respective subscales of positive attitudes, negative attitudes, and subjective norms toward plagiarism.

To interpret the descriptive findings, mean scores obtained from the 5-point Likert-type scales were categorized as follows: 1.00–1.79=very low, 1.80–2.59=low, 2.60–3.39=moderate, 3.40–4.19=high, and 4.20–5.00=very high. These categories were used throughout the analysis to label the levels of AI literacy and attitudes toward plagiarism.

Data Analysis

The first two research questions were addressed by calculating descriptive statistics, means, and standard deviations, for each item on the scales used in the survey. In answering the third question, an analysis of variances (ANOVA) was used to determine if AI literacy and attitudes toward plagiarism differed based on demographic variables. Prior to ANOVA, the assumptions of independence of data and homogeneity of variances were to be met (Field, 2018). Also addressing the third question in terms of gender and study year differences, independent samples *t*-tests were run. As for the fourth research question, a Pearson's correlation test was employed under the assumptions of normality, linearity, and homoscedasticity (Field, 2018; Tabachnick & Fidell, 2019).

To assess the assumptions, both skewness and kurtosis values were examined, as well as Q-Q plot analysis for the assumption of normality for the three measures: independent samples *t*-test, ANOVA, and Pearson's correlation (Field, 2018). Both skewness and kurtosis values were in the range of ± 1 , which indicates normal distribution (Hair et al., 2013). The Q-Q plots also supported the data's normal distribution as illustrated in Fig. 1. As for the assumption of homogeneity of variances, Levene's test for the grouping variable of frequency of AI use indicated there was not a significant violation of the assumption of homogeneity of variances for both scales' mean values [$F_{\text{AILS}}(3, 78)=0.49, p=.68, F_{\text{ATPS}}(3, 78)=1.77, p=.15$], so ANOVA was run normally.

Limitations

Although the present study adopted a structured quantitative design with validated instruments and rigorous analysis, certain limitations should be acknowledged. First, the use of self-reported survey data may have introduced bias due to social desirability or limited recall (Podsakoff et al., 2003). Second, the cross-sectional design limited the interpretation of developmental trends and causal relationships (Wang & Cheng, 2020). Lastly, the study involved participants from a single ELT department in Turkey, which limits the generalizability of the findings. The homogeneity of the sample may have also contributed to the lack of statistically significant demographic differences.

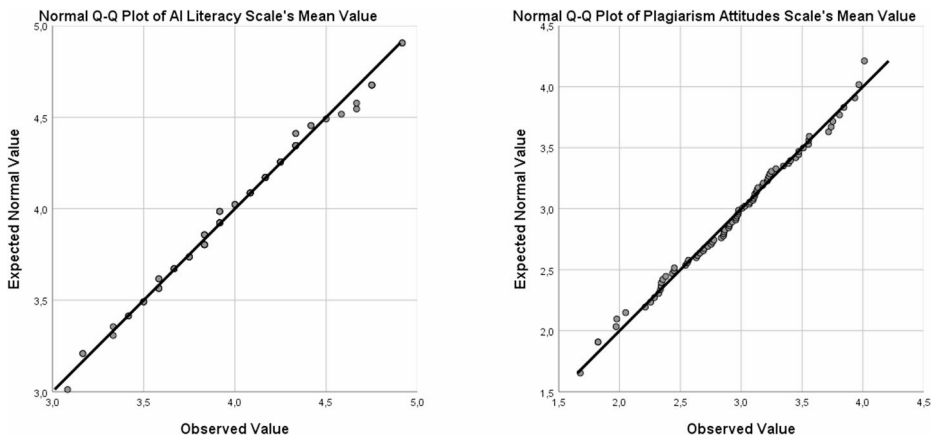


Fig. 1 Normal Q-Q Plots of the Data Mean Values

Table 2 AI literacy levels of participants ($N=82$)

Subscales and Items	M	SD
<i>Awareness</i>	4.26	0.55
AW1. I can distinguish between smart devices and non-smart devices.	4.56	0.59
AW3. I can identify the AI technology employed in the applications and products I use.	4.18	0.77
AW2. I do not know how AI technology can help me. ^R	4.06	1.25
<i>Usage</i>	4.19	0.61
US3. I can use AI applications or products to improve my work efficiency.	4.37	0.65
US1. I can skilfully use AI applications or products to help me with my daily work.	4.37	0.69
US2. It is usually hard for me to learn to use a new AI application or product. ^R	3.85	1.15
<i>Evaluation</i>	4.15	0.56
EV1. I can evaluate the capabilities and limitations of an AI application or product after using it for a while.	4.24	0.57
EV2. I can choose a proper solution from various solutions provided by a smart agent.	4.22	0.64
EV3. I can choose the most appropriate AI application or product from a variety for a particular task.	4.01	0.88
<i>Ethics</i>	3.39	0.63
ET3. I am always alert to the abuse of AI technology.	3.63	1.00
ET1. I always comply with ethical principles when using AI applications or products.	3.39	1.15
ET2. I am never alert to privacy and information security issues when using AI applications or products. ^R	3.16	1.20
Overall Score	4.00	0.40

Items marked with (R) are reverse-coded to ensure interpretive consistency across the scale

Findings and Discussion

Levels of AI Literacy and Plagiarism Attitudes

To address the first and second research questions, the participants' levels of AI literacy and attitudes toward plagiarism were measured with descriptive statistics. As shown in Table 2, the overall mean AI literacy score was high ($M=4.00$, $SD=0.40$), which indicates that EFL

students understood themselves to be competent in understanding and using technologies powered by AI, which is consistent with other similar research reporting on higher education learners with moderate to high levels of AI literacy (Hossain et al., 2025; Xiao et al., 2024). Among the subscales, awareness had the highest mean score ($M=4.26$, $SD=0.55$), indicating a strong recognition of AI technologies as noted in Chan and Hu (2023). However, it is worthwhile to note the slightly lower mean score of the reverse-coded item AW2 indicates that uncertainty persists in understanding what AI is for, or the purposes of AI. This finding again supports the findings of Hossain et al. (2025) which reported students often relied on using the tools or applications without a deeper conceptual understanding of the underlying principles of the technology.

Next, the mean score of the *usage* subscale indicated a high degree of confidence for students using AI technologies ($M=4.19$, $SD=0.61$), especially in regard to using AIs for productivity, consistent with Zhang et al. (2025). However, the lower score on US2 and similar evidence from Kajiwaru and Kawabata (2024) indicate that students still faced challenges when the tool was new or unclear, indicating that use may be context specific or task-based rather than generalizable competence.

Students also scored high for the *evaluation* subscale ($M=4.15$, $SD=0.56$) noting their comfort level with evaluating AI tools. Du et al. (2024) supported this statement, but the findings of Hossain et al. (2025) and Chan and Hu (2023) raise questions about students' accuracy of evaluating tools with respect to accuracy or bias, indicating a potentially over-inflated perception of their skills. Thus, even though students claim they are competent in evaluating AI tools, results may indicate evaluation skills may be more superficial than students believed.

The *ethics* subscale received the lowest mean score with a moderate level of understanding in AI ethics ($M=3.39$, $SD=0.63$). This suggests that participants had limited awareness and ability to navigate key issues such as data privacy and responsible AI use, which aligns with findings from similar studies (Chiu et al., 2024; Kajiwaru & Kawabata, 2024). While these results reflect a general lack of ethical competence, some scholars offer differing perspectives on how such awareness can be improved. For instance, Du et al. (2024) proposed that students' ethical awareness may gradually increase through more frequent and general exposure to AI technologies. However, Kovári (2024) emphasized that meaningful development in AI ethics is unlikely without explicit instructional support. Taken together, these findings reinforce the need to foster comprehensive AI literacy, which not only includes functional skills but also ethical awareness. Addressing this broader competency is essential for preparing future EFL teachers to use AI responsibly and effectively in educational settings.

In Table 3, participants' attitudes toward plagiarism are presented. The participants' plagiarism attitudes were at a moderate level ($M=2.93$, $SD=0.52$), suggesting a more complex view rather than a negative attitude overall.

Of the subscales, *negative attitudes* had the highest mean with a moderate score ($M=3.26$, $SD=0.60$) and indicated a generic awareness around the seriousness of plagiarism as an ethical violation. Szabo and Underwood (2004) have also shown that students with stronger moral reasoning are less likely to assert and idea of academic dishonesty to be acceptable. Similarly, Bairmani et al. (2021) and Çiftçi Birincibubar (2023) determined that students conceptually dislike plagiarism, but few functionally understand the specifics of scholarly

Table 3 Participants' attitudes toward plagiarism

Subscales and Items	M	SD
<i>Positive Attitudes</i>	2.92	0.74
PA3. Self-plagiarism is not punishable because it is not harmful (one cannot steal from oneself).	3.72	1.21
PA5. Self-plagiarism should not be punishable in the same way as plagiarism is.	3.65	1.25
PA1. Sometimes one cannot avoid using other people's words without citing the source, because there are only so many ways to describe something.	3.62	1.02
PA2. It is justified to use previous descriptions of a method, because the method itself remains the same.	3.59	0.87
PA6. Young researchers who are just learning the ropes should receive milder punishment for plagiarism.	3.29	1.23
PA12. If a colleague of mine allows me to copy from her/his paper, I'm NOT doing anything bad, because I have his/her permission.	2.91	1.44
PA10. When I do not know what to write, I translate a part of a paper from a foreign language.	2.72	1.30
PA9. Short deadlines give me the right to plagiarize a bit.	2.55	1.36
PA4. Plagiarized parts of a paper may be ignored if the paper is of great scientific value.	2.52	1.35
PA11. It is justified to use one's own previously published work without providing citation in order to complete the current work.	2.28	1.23
PA7. If one cannot write well in a foreign language (e.g., English), it is justified to copy parts of a similar paper already published in that language.	2.27	1.12
PA8. I could not write a scientific paper without plagiarizing.	1.98	1.19
<i>Negative Attitudes</i>	3.26	0.60
NA3. In times of moral and ethical decline, it is important to discuss issues like plagiarism and self-plagiarism.	4.05	0.96
NA4. Plagiarizing is as bad as stealing an exam.	3.79	1.21
NA2. The names of the authors who plagiarize should be disclosed to the scientific community.	3.52	1.10
NA5. Plagiarism impoverishes the investigative spirit.	3.51	1.12
NA1. Plagiarists do not belong in the scientific community.	3.45	1.24
NA7. Since plagiarism is taking other people's words rather than tangible assets; it should NOT be considered as a serious offense.	2.30	1.13
NA6. A plagiarized paper does no harm to science.	2.22	1.16
<i>Subjective Norms</i>	2.60	0.81
SN2. Those who say they have never plagiarized are lying.	3.45	1.12
SN1. Authors say they do NOT plagiarize, when in fact they do.	3.40	1.00
SN7. Sometimes I copy a sentence or two just to become inspired for further writing.	2.88	1.32
SN8. I don't feel guilty for copying verbatim a sentence or two from my previous papers.	2.78	1.46
SN3. Sometimes I'm tempted to plagiarize, because everyone else is doing it (students, researchers, physicians).	2.60	1.23
SN5. I work (study) in a plagiarism-free environment.	2.50	1.11
SN9. Plagiarism is justified if I currently have more important obligations or tasks to do.	2.28	1.24
SN10. Sometimes, it is necessary to plagiarize.	2.22	1.31
SN6. Plagiarism is not a big deal.	2.11	1.16
SN4. I keep plagiarizing because I haven't been caught yet.	1.85	1.09
Overall Score	2.93	0.52

misconduct across different forms, including the murkier allegations, such as self-referencing, and the matter of paraphrasing.

The positive attitudes subscale, with a moderate mean score, climbed into partial acceptance of practices like self-plagiarism and technical or methodology content reuse ($M=2.92$, $SD=0.74$). Both PA3 and PA5 attained relatively significant ratings suggesting that students regarded self-plagiarism as less serious. This is consistent with Kaler et al. (2023) stating that undergraduate students distinguish self-plagiarism from other kinds of academic misconduct and that they consider self-plagiarism to be a less serious form of academic misconduct. Similarly, Çiftçi Birincibubar (2023) reported that similar results in which post-graduate EFL students expressed their uncertainty and lenience upon their self-copying and that self-copying did not elicit ethical scrutiny. Both studies imply that self-copying is an area in which students may not initiate ethical sensitivity.

The subjective norms subscale had a moderate mean score as well ($M=2.60$, $SD=0.81$) and was ambiguous in signifying peer and/or institutional perspectives about plagiarism. While students did not generally support the normalization of plagiarism (i.e., low means for SN6 and SN4), the higher scores for items like SN1 and SN2 indicated some ambivalence or scepticism for other people's commitment to academic honesty. This is consistent with Birincibubar's (2023) claim that students perceive a disconnect or hypocrisy in academic communities' attitudes toward plagiarism. Likewise, Muluk et al. (2021) determined that, while peers may compete, family obligations may also pressure students, where some students feel justified in behaving dishonestly provided it meets the standards of performance.

Further supporting these findings, Doró (2014) and Muluk et al. (2021) identified the factors of time constraints, an elevated academic workload, and misinterpretation of paraphrasing or writing in academic English as regular rationalizations for plagiarism among EFL students. Consequently, these contextual pressures, with accessible online content and inadequate training in citation referencing, may encourage students to plagiarize even when they recognize the ethical issues behind it. Mostly, the findings also indicate an interaction between ethical awareness, contextual challenges, and social factors impacting individuals' conceptualizations of plagiarism and responses.

Demographic Differences Across the Variables

Addressing the third research question, the study sought to determine demographic differences across the variables. Independent samples t-tests revealed no statistically significant difference in AI literacy levels between the second-grade ($M=3.97$, $SD=0.49$) and the fourth-grade students ($M=4.03$, $SD=0.34$), $t(62.09) = -0.60$, $p=.55$. Similarly, plagiarism attitudes did not significantly differ between the two groups ($M_{2nd}=2.95$, $SD_{2nd}=0.51$; $M_{4th}=2.91$, $SD_{4th}=0.54$) either, $t(80)=0.42$, $p=.67$. These findings are in line with O'Dea et al. (2024), who found academic grade-level had no significant impact on AI literacy among university students across the UK and Hong Kong. One possible explanation is that both year groups in the present study had comparable exposure to AI technologies and similar access to academic integrity policies, as digital competencies and ethical considerations are integrated into the curriculum of the institution.

Comparisons by gender again did not demonstrate any statistically significant results. Male ($M=3.98$, $SD=0.35$) and female students ($M=4.03$, $SD=0.47$) had AI literacy scores that were nearly identical, $t(80) = -0.55$, $p=.58$, and attitudes toward plagiarism were virtu-

Table 4 Descriptive statistics of frequency of AI use

Frequency of AI Use	AI Literacy		Plagiarism Attitudes	
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
Rarely	3.54	0.53	3.00	0.19
Sometimes	3.94	0.47	2.79	0.56
Often	4.00	0.38	2.98	0.46
Always	4.13	0.38	2.96	0.67

The category of “never” was not included due to its frequency being 0 in numbers

Table 5 ANOVA results of the variables based on frequency of AI use

Dependent Variable	Source	SS	df	MS	F	<i>p</i>
AI Literacy	Between groups	0.78	3	0.26	1.59	0.19
	Within groups	12.74	78	0.16		
	Total	13.52	81			
Plagiarism Attitudes	Between groups	0.46	3	0.15	0.55	0.64
	Within groups	22.01	78	0.28		
	Total	22.48	81			

SS=Sum of Squares; *MS*=Mean Square; *F*=F-ratio; *p*=significance level

ally the same (both $M=2.93$, $SD_{\text{male}}=0.48$, $SD_{\text{female}}=0.58$), $t(80)=0.04$, $p=.96$. These findings align with previous studies that reported no significant gender differences in AI literacy (e.g., Mansoor et al., 2024) or attitudes toward plagiarism (Chen, 2022; Syed Ahmad et al., 2022). However, studies such as Atikuzzaman and Ahmed (2025) did report gender-related differences in perceptions of plagiarism, suggesting that some students viewed female peers as less likely to engage in misconduct. Such findings, when observed, may be shaped by sociocultural norms or institutional environments that influence gendered expectations and student behaviour. Therefore, the presence or absence of demographic differences in this context should be interpreted in light of broader cultural or institutional factors that vary across settings.

As for the extent to which frequency of AI use had an impact on students’ AI literacy or plagiarism attitudes, one-way ANOVA was performed with results provided in Tables 4 and 5.

While AI literacy scores were generally increasing with frequency of use from an average of 3.54 ($SD=0.53$) for infrequent users to 4.3 ($SD=0.38$) for frequent users, the means did not reach statistical significance, $F(3,78)=1.59$, $p=.19$. Likewise, there were no significant differences in plagiarism attitudes, $F(3,78)=0.55$, $p=.64$. These results are consistent with Chan (2023) who found that AI usage frequency was positively related to AI competence, but they also noted that increased frequency may not lead to significant group differences in some studies, especially when groups are close in their level of educational access and exposure. In this study, the lack of significance in these areas may demonstrate the general normalisation of AI use as felt among the student population, thus underlining the diminishing impact of usage frequency as a differentiating factor.

Overall, the lack of significant demographic and usage-based differences across AI literacy and plagiarism attitudes may be attributed to shared educational experiences, consistent curricular exposure, and widespread access to AI tools among EFL students. These results reinforce the need to go beyond demographic targeting in AI or integrity education

and instead focus on universal, ethics-integrated instruction that addresses the nuanced challenges faced by all learners.

Mismatch between AI Literacy and Plagiarism Attitudes

Finally, in answering the fourth research question to explore the potential relationship between student AI literacy levels ($M=4.00$, $SD=0.40$) and student attitudes toward plagiarism ($M=2.93$, $SD=0.52$), a Pearson's correlation test was conducted. The results demonstrated a small, negative, but statistically non-significant correlation between the two variables, $r(82)=-0.17$, $p=.137$, indicating no noteworthy relationship between how literate students felt with AI and how ethically permissive they felt toward plagiarism.

The correlation was expectedly negative, implying that, as student AI literacy increased, students may have felt slightly less ethically permissive toward plagiarism, but the relationship was weak and not statistically significant. Therefore, higher AI literacy does not necessarily translate into heightened ethical awareness (Chiu et al., 2024; Kajiwara & Kawabata, 2024). Likewise, as in Hossain et al. (2025), many students using AI tools for writing in academia still lack a critique of ethical considerations such as originality, authorship, and responsible use. Supporting these, some research has suggested that ethical reasoning is a component of AI literacy and requires explicit instruction and reflection rather than being a development that happens automatically with further use of AI tools (Du et al., 2024; Kovári, 2024).

Conclusion and Recommendations

This research sought to investigate the correlation between AI literacy and attitudes toward plagiarism among ELT students in an EFL writing context, with an emphasis on demographic variables, as well as frequency of AI usage. Participants demonstrated a generally high level of AI literacy, but their understanding of ethical considerations lagged behind. This was particularly seen in the issues of privacy, responsible AI use, and authorship. The participating students' attitudes regarding plagiarism also offered a mixed picture. They agreed on the seriousness of academic misconduct but also participated to some degree in self-plagiarism and reuse of technical content. This situation suggests some degree of mild indifference to misconduct.

Next, the lack of a statistically significant correlation between AI literacy and attitudes toward plagiarism indicates that students may have developed functional competence with regard to AI technologies, but that competence does not necessarily extend to the ethical use of AI. In other words, students may feel comfortable using and assessing the effectiveness of AI technologies, but that comfort does not imply that they have an understanding of originality, authorship, or the ethical use of AI (Zhai et al., 2024). It suggests, as presented in the literature, that ethical reasoning may have to be taught directly, rather than passively learned through exposure to AI technologies. This is particularly meaningful as the use of AI technologies become normalized in academia, raising the stakes for academic programs regarding whether the teaching of ethics will take precedence over the use of these technologies.

Likewise, the lack of statistically significant differences among demographic variables such as year in academic program, gender, and frequency of AI use suggest a homogenizing

effect related to the institution. When all students share similar curricula, academic policies, and technology, their experiences and attitudes toward the use of AI may have likely converged, minimizing individual variance. This suggests that institutional structure and pedagogy have a larger role in determining students' digital and ethical literacies. For ELT students, who will soon be entering roles as educators, this finding highlights the need to integrate ethical digital literacy education consistently across all preservice teacher profiles. Such efforts should not only focus on what it means to be a teacher today but also prepare them to guide future generations of learners.

Although modest in sampling size and context-specific, the present study provides valuable insights. It highlights consistent themes which correspond with the existing literature, specifically about students' operational AI knowledge and the disconnection of their ethical reasoning about tool use (Chiu et al., 2024; Hossain et al., 2025; Kajiwaru & Kawabata, 2024). The present research underscores a structural issue regarding how AI literacy is constructed in teacher education programs. Rather than assuming ethical reasoning is a natural extension of being using new technological tools, the literature calls for explicit opportunities to integrate ethical engagement through careful curriculum and pedagogical design (Du et al., 2024; Kovári, 2024). Thus, this study provides relevant data which can be used as a source for informing a variety of instructional practices and curricular decisions in ELT and teacher education contexts, especially in the age of AI where educational contexts and uses are continually evolving.

The future of education is undoubtedly connected to AI, and teaching technological competence is no longer enough. AI literacy, in this regard, needs an ethical and critical dimension, especially for preservice teachers, who will influence students' digital behaviours and values. The findings of this study reaffirm the need for teacher education programs to conduct explicit training on AI ethics and academic integrity. This could help to ensure that future educators are not only competent users of AI, but advocates for ethical, transparent, and responsible use in the changing landscape of language education.

Building on these findings, the study offers practical implications for ELT curriculum designers and teacher educators aiming to promote ethical AI use. Specifically, targeted training modules can be developed to simulate real-world dilemmas involving AI in educational settings, such as using AI for assignment completion, misattributing authorship, or handling student data privacy. These may include case-based learning, scenario analysis, and guided reflective discussions, all of which have been shown to foster critical thinking and moral reasoning in preservice teachers (Kovári, 2024; Du et al., 2024). Rather than being isolated in stand-alone seminars, such modules should be embedded throughout the ELT curriculum in courses like academic writing, materials development, and instructional technologies, allowing sustained engagement and deeper transfer of ethical reasoning to future teaching practices (Chiu et al., 2024; Hossain et al., 2025). In addition to curricular design, institutions may also consider implementing clear policy frameworks on AI usage, embedding academic integrity statements specifically addressing AI-generated content, and organizing faculty-led discussions on AI ethics. Although these strategies are yet to be widely examined in empirical research, they represent forward-looking institutional efforts to cultivate a culture of academic responsibility and equip future educators to model ethical digital behaviours.

Lastly, although the study's methodological limitations, such as the use of self-reported data, a single-institution sample, and a cross-sectional design, have already been addressed

in the methodology section, they need to be briefly reiterated here as they may have limited the generalizability and causal interpretability of the findings (Podsakoff et al., 2003; Wang & Cheng, 2020). Future research could address these limitations by incorporating behavioural measures alongside self-reports, employing longitudinal designs to observe change over time, and including broader, more diverse samples. Moreover, intervention-based studies focusing on AI ethics training could offer valuable insights into how students' ethical awareness and academic integrity evolve in response to targeted instruction.

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Author Contributions Emre Uygun solely contributed to the conceptualization and design of the study, data collection, analysis, and interpretation, as well as drafting, revising, and approving the final manuscript.

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Data Availability The data that support the findings of this study are not publicly available due to privacy concerns and the terms of participant consent.

Declarations

Ethics Approval The study was approved by the Ethical Review Board of Çanakkale Onsekiz Mart University's School of Graduate Studies (Date: 16.05.2025, Issue: E-80110942-050.99-2500129211, Number: 114/172).

Consent All participants provided informed consent prior to participation. The consent form was included on the first page of the research instruments. It informed participants about the study's purpose, voluntary nature, anonymity of responses, and agreement for the publication of anonymized findings.

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